

WASP-11b

R-0663

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_260110 · created 2026-07-27T17:49:01+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461050.6774 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.132 +/- 0.01
Transit depth (Rp/Rs)^2	1.73 +/- 0.27 [%]
Orbital Inclination (inc)	89.98 +/- 0.52
Airmass coefficient 1 (a1)	1.122 +/- 0.01
Airmass coefficient 2 (a2)	-0.098 +/- 0.016
Scatter in the residuals of the lightcurve fit is	0.94 %
Best Comparison Star	None
Optimal Aperture	4.6
Optimal Annulus	7.03
Transit Duration (day)	0.1068 +/- 0.0013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.132 ± 0.01 vs 0.131 (deviation 0.1σ)
Duration fit vs published	0.1068 d vs 0.1075 d (deviation 0.54σ)
Mid-time O–C	0.3 min
Depth SNR	13.2
Residual scatter	0.94 %

Light curve

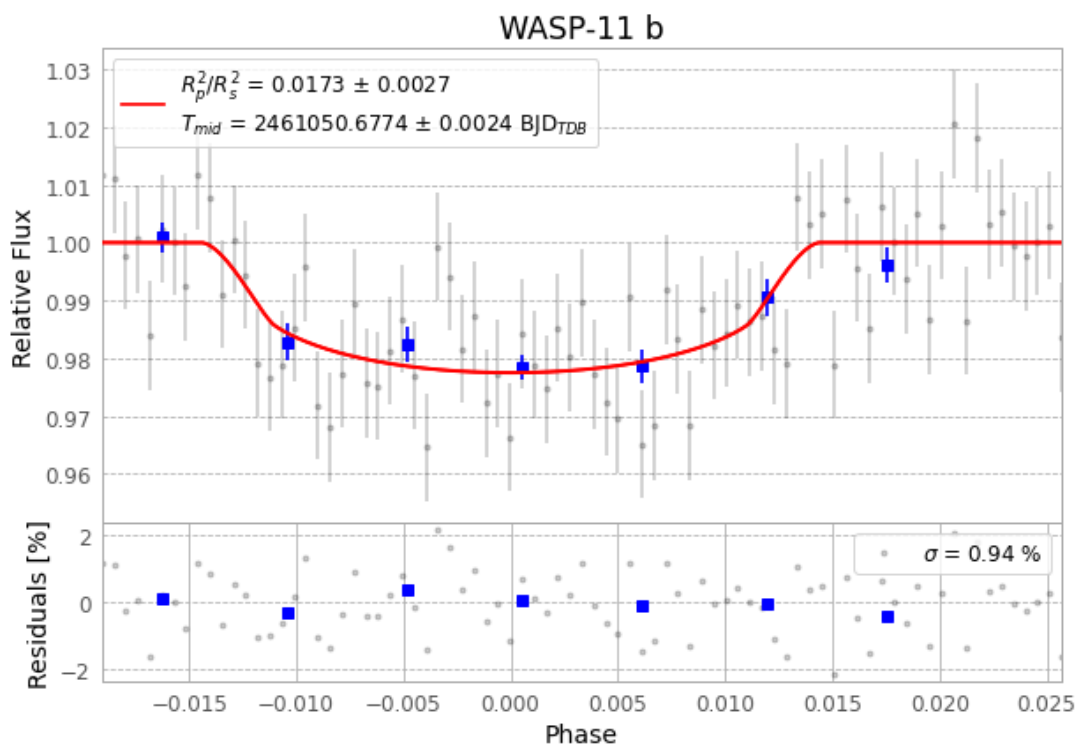


Figure 1 — FinalLightCurve_WASP-11 b_2026-01-09.png (SHA-256 8ea4d9244ed05830...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 324], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 81b50afcf8aeb05c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 574e53e

Data provenance

81 FITS frames (53 MB), HATP-10260110022716.FITS → HATP-10260110062716.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-260110.log (45b64bedf9fc...), FinalLightCurve_WASP-11 b_2026-01-09.png (8ea4d9244ed0...), FinalLightCurve_WASP-11 b_2026-01-09.csv (0f7af96338b1...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 17:49Z.